INCH-POUND

MIL-C-24643/56C
14 March 1994
SUPERSEDING
MIL-C-24643/56B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWL-7
(INCLUDING VARIATION LS2SWLA-7)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2SWL-7 (unarmored) and LS2SWLA-7 (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, AWG size 16, class B stranding.

* Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.019 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.

Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form a pair.

Fifth - Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.

* Sixth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

AMSC N/A FSC 6145
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Construction (watertight) (Continued)

* Seventh- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

* Eighth - Seven shielded pairs cabled together with a lay not greater than 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well-rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS2SWL-7

* Tenth - Cross-linked polyolefin jacket. Minimum average jacket thickness 0.075 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum inch.

Construction variant for type LS2SWLA-7

* Tenth - Cross-linked polyolefin jacket. Minimum average jacket thickness 0.075 inch. Cable overall diameter shall be 0.910 minimum, 0.960 maximum inch.

* Eleventh - Braided metal armor.

EXAMINATION AND TESTS:

| Basic electrical: | Requirements |
|---|---|
| Conductor Resistance - ohms/1000 feet at 25°C, maximum | 4.66 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2000 |
| Conductor to shield | 1000 |
| Shield to shield (pair shields only) | 500 |
| Insulation resistance - Megohms/1000 feet minimum | |
| * Conductor to shield | 500 |
| * Component shield to shield | 100 |
| Component and shield continuity | No failure |

| Group A: | |
|---|---|
| Visual and dimensional | No failure |
| Watertightness - see MIL-C-24643 for water leakage | No failure |

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| EXAMINATION AND TESTS (Continued) | Requirements |
|---|---|
| Capacitance | |
| Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum | 35 |
| Capacitance unbalance - percent maximum | 10 |
| Characteristic impedance - at 1MHz, ohms | 75 ±5 |
| Group B: | |
| * Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200 °C) | 50 |
| Drip - 95 ±1°C | Zero |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength - lb/in$^2$, minimum | 1800 |
| Elongation - percent, minimum | 250 |
| Jacket (cable) | |
| Tensile strength - lb/in$^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Group C: | |
| Physicals - (aged) | |
| Insulation | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged minimum | 80 |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged minimum | 60 |
| Hot oil immersion | |
| * Tensile strength - percent of unaged, minimum | 50 |
| * Elongation - percent of unaged, minimum | 50 |
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 30 |
| * Permanence of printing (jacket) - (type LS2SWL-7 only) cycles, minimum | 125 |
| * Cable sealant removability | No failure |
| Armor - (Type LS2SWLA-7 only) - conformance to material, construction and coverage | No failure |
| Shield conformance to material, construction and coverage | No failure |



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EXAMINATION AND TESTS (Continued) Requirements

Group D:
Flame propagation (cable)..... No failure

QUALIFICATION INSPECTION:
* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

| | Requirements |
|---|---|
| Aging and compatibility (cable) (125 ±5°C.)..... | No failure |
| Abrasion resistance (jacket) - scrapes, minimum..... | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket..... | 2 |
| Fillers..... | 2 |
| Insulation..... | 18 |
| Halogen content - percent, maximum | |
| Jacket..... | 0.2 |
| Fillers..... | 0.2 |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum..... | 50 |
| Elongation - percent of unaged, minimum..... | 50 |
| Smoke index, maximum | |
| Jacket..... | 25 |
| * Fillers..... | 45 |
| Insulation..... | 45 |
| Toxicity index, maximum | |
| Jacket..... | 5 |
| Fillers..... | 5 |
| Insulation..... | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum..... | 80 |
| Weathering (jacket)..... | No failure |

UNIT ORDERING LENGTHS: 1500 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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